

File permissions in Linux

Project description

The purpose of this project is to demonstrate my experience using Linux commands in bash to manage file permissions by providing examples of various commands, demonstrating how they are executed and describing their functions as related to managing file permissions.

Check file and directory details

We will start with an overview of the file structure of the `/home/researcher2/projects` directory and the permissions of the files and subdirectory it contains. To do this we will open a bash terminal window and enter the command `cd /home/researcher2/projects`. This will instruct the computer to navigate to the destination of the directory we will be working with.

```
researcher2@cb92fcef663c:~$ cd /home/researcher2/projects
researcher2@cb92fcef663c:~/projects$
```

The task is to examine and manage the permissions of the files in the `/home/researcher2/projects` directory for the `researcher2` user. In order to examine the files and subdirectories contained within our current working `/home/researcher2/projects` directory as well as all the permissions we would enter the command `ls -la` from the current working directory. This command instructs the computer to display all files and subdirectories contained within the `/projects` directory as well as all permissions for each

```
researcher2@cb92fcef663c:~$ cd /home/researcher2/projects
researcher2@cb92fcef663c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 00:49 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:29 ..
-rw--w---- 1 researcher2 research_team  46 Nov 21 00:49 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 00:49 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Nov 21 00:49 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 00:49 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 00:49 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 00:49 project_t.txt
researcher2@cb92fcef663c:~/projects$
```

In the example above we see the output of the command we executed, It displays five files with the following names and permissions:

- `project_k.txt`
 - User = read, write,
 - Group = read, write
 - Other = read, write
- `project_m.txt`
 - User = read, write
 - Group = read
 - Other = none
- `project_r.txt`
 - User= read, write
 - Group = read, write
 - Other = read
- `project_t.txt`
 - User = read, write
 - Group = read, write
 - Other = read
- `.project_x.txt`
 - User = read, write
 - Group = write
 - Other = none

There is also one subdirectory inside the `projects` directory named `drafts`. The permissions on `drafts` are:

- User = read, write, execute
- Group = execute
- Other = none

Describe the permissions string

The Permissions string as shown above is a 10-character string that can be deconstructed to determine who is authorized to access each individual file and their specific permissions. The characters and what they represent are as follows:

- 1st character: This character is either a `d` or hyphen (`-`) and indicates the file type. If it's a `d`, it's a directory. If it's a hyphen (`-`), it's a regular file.

- 2nd-4th characters: These characters indicate the read (`r`), write (`w`), and execute (`x`) permissions for the user. When one of these characters is a hyphen (`-`) instead, it indicates that this permission is not granted to the user.
- 5th-7th characters: These characters indicate the read (`r`), write (`w`), and execute (`x`) permissions for the group. When one of these characters is a hyphen (`-`) instead, it indicates that this permission is not granted for the group.
- 8th-10th characters: These characters indicate the read (`r`), write (`w`), and execute (`x`) permissions for other. This owner type consists of all other users on the system apart from the user and the group. When one of these characters is a hyphen (`-`) instead, that indicates that this permission is not granted for other.

In our example, the file permissions for `project_t.txt` are `-rw-rw-r--`. Since the first character is a hyphen (`-`), this indicates that `project_t.txt` is a file, not a directory. The second, fifth, and eighth characters are all `r`, which indicates that user, group, and other all have read permissions. The third and sixth characters are `w`, which indicates that only the user and group have write permissions. No one has execute permissions for `project_t.txt`.

Change file permissions

Next we are going to discuss how to change permissions on files, as an example, We are going to change file permissions for other so they don't have write access to any of their files. To comply with this, I referred to the file permissions that I previously returned and determined `project_k.txt` must have the write access removed for other. To do this I will enter the command `ch mod o-w project_k.txt`

The following example demonstrates how I used Linux commands to do this:

```

researcher2@508326222e30:~$ cd /home/researcher2/projects
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-rw--w---- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ chmod o-w project_k.txt
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-rw--w---- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ █

```

The first two lines of the screenshot display the commands I entered, and the other lines display the output of the second command. The `chmod` command changes the permissions on files and directories. The first argument indicates what permissions should be changed, and the second argument specifies the file or directory. In this example, I removed write permissions from other for the `project_k.txt` file. After this, I used `ls -la` to review the updates I made.

Change file permissions on a hidden file

Now we will explore how to change file permissions on a hidden file. Using our previous example the file `project_x.txt` has been archived. To demonstrate how to change file permissions on a hidden file I will remove write access to this project, but still allow user and group to have read access. To show how this is done first I will enter the command `chmod u-w,g-w,g+r .project_x.txt` to change the file permissions and then enter the command `ls -la` to confirm that the changes have been applied

```

researcher2@508326222e30:~$ cd /home/researcher2/projects
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-rw--w---- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ chmod o-w project_k.txt
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-rw--w---- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-r--r----- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ █

```

The first two lines of the screenshot display the commands I entered, and the other lines display the output of the second command. The file `.project_x.txt` is a hidden file because it starts with a period (`.`). In this example, I removed write permissions from the user and group, and added read permissions to the group. I removed write permissions from the user with `u-w`. Then, I removed write permissions from the group with `g-w`, and added read permissions to the group with `g+r`. Finally, I instructed the computer to display the updated permissions by entering the command `ls -la` once again.

Change directory permissions

Lastly we will go over how to change directory permissions. Using the example we have been working with, I will modify `researcher2` user to have access to the `drafts` directory and its contents. This means that no one other than `researcher2` should have execute permissions. I will do so by entering the command `chmod g-x drafts` followed by the `ls -la` command to

once again display all user permissions assigned in every file, hidden file and subdirectory that is contained in the `/projects` directory

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-rw--w---- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ chmod o-w project_k.txt
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-rw--w---- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-r--r----- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$ chmod g-x drafts
researcher2@508326222e30:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 01:55 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov 21 02:30 ..
-r--r----- 1 researcher2 research_team  46 Nov 21 01:55 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Nov 21 01:55 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Nov 21 01:55 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Nov 21 01:55 project_t.txt
researcher2@508326222e30:~/projects$
```

As you can see in this example, the output displays the permission listing for several files and directories. Line 1 indicates the current directory (`projects`), and line 2 indicates the parent directory (`home`). Line 3 indicates a regular file titled `.project_x.txt`. Line 4 is the directory (`drafts`) with restricted permissions. Here you can see that only `researcher2` has execute permissions. It was previously determined that the group had execute permissions, so I used the `chmod` command to remove them. The `researcher2` user already had execute permissions, so they did not need to be added.

Summary

I changed multiple permissions to match the level of authorization my organization wanted for files and directories in the `projects` directory. The first step in this was using `ls -la` to check the permissions for the directory. This informed my decisions in the following steps. I then used the `chmod` command multiple times to change the permissions on files and directories.